Miniature Soft Vine Robots for Transcervical Ultrasound Imaging

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***Abstract*—Minimally invasive cervical navigation remains challenging due to constrained anatomy and limitations of rigid instruments. This work presents a 5 mm soft vine robot integrated with an intravascular ultrasound (IVUS) probe for cervical navigation and imaging. The system navigation was evaluated in a female reproductive system phantom and ex vivo ovine tissue. The robot successfully navigated the cervical canal at 35 kPa growing pressure, with lower insertion forces than a rigid probe (0.15 N vs. 3.22 N). IVUS measurements captured the overall lumen geometry with a mean bias of 0.65 $\pm$ 0.06 mm, corresponding to a mean percentage error of 12.8 $\pm$ 1.2%. These results demonstrate the potential of soft vine robots for low-force, image-guided access through the cervix into the uterus.**

***Index Terms*—Soft Sensors and Actuators, Soft Robot Materials and Design, Surgical Robotics: Steerable Catheters/Needles**

I. INTRODUCTION

Minimally invasive access to the uterus via the cervix remains a persistent challenge in gynecology [1]. Current endoluminal methods rely on rigid hysteroscopes or dilators that force passage through the cervix, a narrow, compliant, and sensitive structure. These probes offer limited maneuverability and an inability to conform to the delicate anatomy. Consequently, they can cause discomfort to the patient and hinder the direct visualization required for early diagnosis of conditions such as endometriosis, adenomyosis, and cancer [2].

Soft growing robots, often referred to as "vine robots", navigate via tip extension driven by internal fluid pressure. Their unique kinematics allow them to traverse narrow lumens and squeeze through gaps of varying diameters, minimizing shear forces along their length [3], [4]. When used as medical catheters, they reduce tissue interaction forces, thereby enhancing safety during interventions [5], [6]. Nonetheless, their use in cervical applications remains largely unexplored. Furthermore, although vine robots have been equipped with sensing modalities such as cameras [7], proprioceptive [8], tactile [9], pressure [10], or curvature sensors [11], the integration of imaging systems remains limited, particularly for medical applications requiring real-time assessment of the surrounding anatomy.

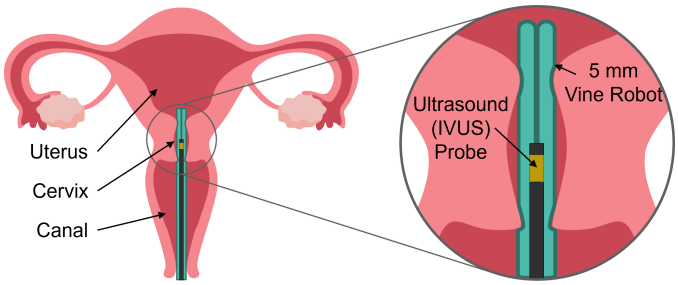


Fig. 1. Transcervical deployment of a 5 mm vine robot integrated with an ultrasound (IVUS) probe.

In this work, we present a millimeter-scale, water-driven, soft everting vine robot with integrated ultrasound imaging (Fig. 1). The design and fabrication strategies are introduced for integrating an Intravascular Ultrasound (IVUS) probe as the tail of a 5 mm vine robot, using a vascular imaging catheter for cervical imaging. The navigation is evaluated in both a female reproductive system phantom and ex vivo ovine tissue. The deployment forces of the soft everting robot are experimentally characterized and compared with those of a

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conventional rigid probe. Finally, experimental results in the female reproductive system phantom demonstrate the successful navigation of the robot in the tight lumen while collecting ultrasound images. Our work establishes the feasibility of ultrasound-integrated vine robots for navigation and imaging within the cervix, building a step towards minimally invasive gynaecological diagnostics.

II. DESIGN AND FABRICATION

The 5 mm vine robot was fabricated from two 38 µm thickness Thermoplastic Polyurethane (TPU) films (Stretchlon 200, Airtech Advanced Materials, UK). A heat press (Combo Heat Press, Creworks, USA) was used to soft-seal the layers for laser cutting (Falcon 2 Pro 22 W, Creality, China) the tubular shape of the robot body, as in previous work [12].

The phantom molds were generated in computer-aided design (CAD) from a GrabCAD geometry (Fig. 2 inset) scaled to anatomical dimensions, then cast in Dragon Skin^{TM} 10 (Smooth-On Inc, USA) silicone to simulate the geometric and mechanical properties of soft tissue. The tissue in the female reproductive system varies significantly depending on multiple factors, including age, gestational stage, and menstrual cycle [13]. Therefore, the material was selected to replicate the approximate resistance to access the uterus in the higher ranges of stiffness reported (Young's modulus $\sim 10^5$Pa) [13].

An IVUS probe (Visions PV.018 Digital IVUS Catheter, Philips Volcano, USA) was attached to the vine robot as its tail and clamped to a motorized linear rail to provide millimetric displacement control (Fig. 2), following our previous system [14]. The probe replaces the active channel, and tension in its tether regulates the eversion process, preventing the probe from moving independently of the vine robot. To enable ultrasound imaging, this system uses pressurised water within the robot lumen as the driving medium rather than the pressurised air commonly used in pneumatic vine robots. The water additionally provides an essential acoustic coupling medium, minimising acoustic impedance mismatch with soft tissue and preserving ultrasound image quality.

III. METHODS

A. Navigation

In order to evaluate the navigation capability of the 5 mm vine robot through the cervical canal towards the uterus, insertion experiments were conducted in both an anatomically representative female reproductive system phantom and ex vivo ovine tissue to demonstrate feasibility in biological tissue. The vine robot was actuated using a programmable pressure regulator (ITV1010-211BL, SMC, Japan), with the gauge pressure increased from $P_g = 25$ kPa in 5 kPa increments until successful eversion and advancement were achieved. Navigation was visually confirmed in the phantom and validated using an endoscopic camera in ex vivo tissue.

B. Insertion Force

In order to understand the mechanical interaction during insertion, axial insertion forces were measured using a load cell (DBCR Compact S-Beam, Applied Measurements, UK) positioned against the phantom, aligned with the direction of robot advancement. The mechanical interaction of the vine robot was compared with that of a rigid probe ($\varnothing_{max}$ = 5 mm) during insertion through the cervical lumen. For the vine robot insertion was achieved by tip eversion driven by pressurized water, while the tail was advanced by the linear platform. In contrast, the rigid probe was inserted through the cervical lumen by linear translation using the same linear platform (Fig. 2). Five insertion trials were performed for each device, and the peak insertion force and trial-to-trial variability were analysed.

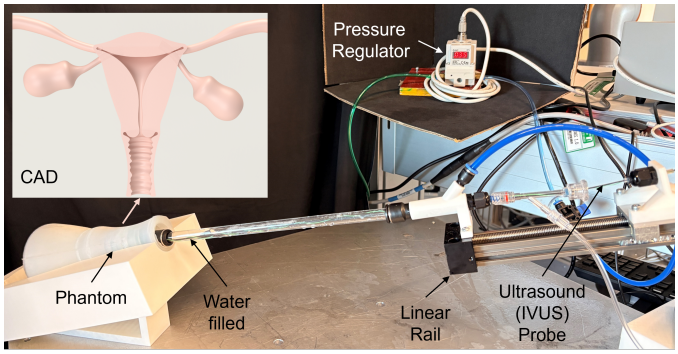


Fig. 2. Experimental setup for vine robot deployment with the IVUS probe into an anatomically scaled phantom of the female reproductive system (inset).

C. Imaging

Radial cross-sectional ultrasound slices were acquired (Volcano CORE Mobile System, Philips Volcano, USA), while the probe was translated at 1 mm intervals along the longitudinal z-axis to a 40 mm insertion depth. The experiment was performed with the robot inserted into the water-filled phantom cervix and repeated five times. Noise was reduced by adjusting the settings for sensitivity, gain, and diameter to 3, 32, and 10 mm, respectively.

The slices were processed using custom Python scripts. Each image was first cropped to the region of interest, with the dead zone ($r \sim 1$ mm) masked out. In each slice, the luminal boundaries were manually identified and outlined, excluding partially visualized cross-sections, and the diameters were subsequently measured using image-processing algorithms based on contour detection. Prior to reconstruction, one representative repetition was aligned based on the centers of the smallest detected diameters. Finally, the segmented contours were stacked at 1 mm voxel resolution to reconstruct the lumen volume.

Measurement accuracy was quantified by comparing the reconstructed diameters with the phantom ground-truth geometry at each (z)-position across all trials. The local bias and percentage error were calculated as:

$$\text{Bias}(z) = D_{\text{measured}}(z) - D_{\text{true}}(z) \tag{1}$$

$$PE(z) = \frac{\text{Bias}(z)}{D_{\text{true}}(z)} \times 100\% \tag{2}$$

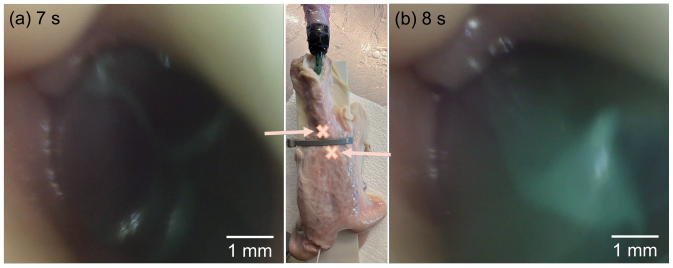


Fig. 3. Endoscopic visualization of robotic deployment during ex vivo tests on ovine cervix and uterus tissue. (a) The tip emerges from the cervix at 7 s. (b) At 8 s, the vine advances $\sim$ 3 mm into the uterus.

where $(D_{\text{true}}(z))$ is the phantom ground-truth diameter at position (z). Bias and percentage error were then averaged over all (z)-positions and trials to obtain the reported mean bias and mean percentage error.

IV. RESULTS

A. Navigation

The initial navigation experiments established that a growing pressure of $P_g = 35$ kPa was required for successful advancement through the reduced cervical lumen, remaining well below the measured burst pressure of 55 kPa and providing a safety margin of 36.4% throughout deployment. At this pressure, the vine robot successfully deployed and advanced through the cervical canal in all five phantom trials. Hence, the same actuation pressure was used for the ex vivo ovine cervix experiments. Fig. 3 shows the successful navigation of the vine robot through the cervix and its advance into the uterine cavity. Endoscopic visualisation confirmed that the 5 mm vine robot reached the uterine cavity, advancing at $\sim$ 3 mm/s.

B. Insertion Force

Fig. 4 compares the insertion forces generated by the vine robot vs the rigid probe during phantom insertion. The vine robot exhibited a maximum insertion force 0.15 $\pm$ 0.01 N and a mean insertion force of 0.02 $\pm$ 0.03 N, compared with 3.22 $\pm$ 0.17 N and 1.00 $\pm$ 0.07 N for the rigid device, respectively. Therefore, across the five trials, the vine robot produced consistently lower insertion forces than the rigid device.

C. Imaging

Fig. 5 illustrates ultrasound-based shape reconstruction of the cervical path within the phantom model. In Fig. 5(a), representative cross-sectional scans show the measured circles at selected insertion depths across the z-axis. The first region, from z = 1 to (i) 12 mm, did not provide sufficient image contrast for reliable circular feature detection. The inner circles from (ii) z = 18 mm through z = 40 mm, including (iii) z = 28 mm, correspond to the vine robot diameter ($\text{Ø} = 5.09 \pm 0.05$ mm). This diameter decreases to $\text{Ø} = 4.56 \pm 0.06$ mm near the end, z = 39 mm.

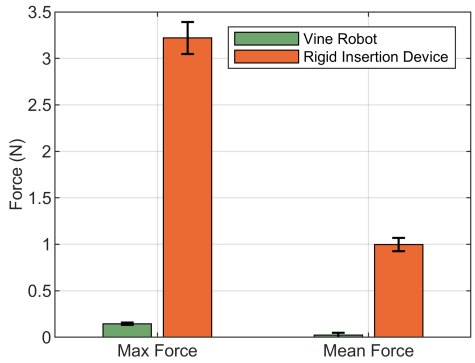


Fig. 4. Comparison of maximum and mean axial insertion forces measured during navigation through the cervical phantom for the vine robot and a rigid probe. Error bars represent the standard deviation across five insertion trials.

Fig. 5(b) shows the sliced longitudinal $y-z$ reconstruction obtained after rotating the image stack 60° clockwise around the z-axis, which provided the most complete cross-sectional representation along the z-axis. Most of the detected luminal edges appeared as fragmented contours due to signal dropout at certain angles. Therefore, Fig. 5(c) plots the two innermost extracted diameters, enabling a direct comparison between the smallest detected circle, denoting the vine, and the phantom profile. The largest diameter is observed around $z = 27$ mm, corresponding to the widest section of the cervical lumen.

Across five repeated IVUS insertions into the phantom, the reconstructed lumen diameter profiles showed high repeatability. Over the range with enough image contrast for reliable circular feature detection (z = 12-40 mm), the inter-trial standard deviation averaged 0.14 mm, corresponding to a coefficient of variation below 3%, indicating that the manual boundary identification and automated contour-based diameter measurement produced consistent diameter estimates across repeated acquisitions. Slightly greater variability was observed around the max diameter ($z = 27$ mm), reaching a maximum inter-trial standard deviation of $\sim$ 0.26 mm.

Compared with the phantom ground truth, the reconstructed diameter profiles consistently overestimated the lumen size by 0.65 $\pm$ 0.06 mm (mean ± SD, N=5), corresponding to a mean percentage error of 12.8 $\pm$ 1.2% of true diameter. This error pattern was systematic rather than caused by random measurement noise, consistent with the low variability observed across repeated acquisitions ($\pm$ 0.06 mm). However, the offset was not uniform along the length. The magnitude of this deviation varied by position (SD = 0.43 mm, range 0.12–1.42 mm across z), with the smallest deviations occurring near the widest section of the phantom (around $z = 27$ mm), despite this being the location of greatest inter-trial variability, whereas the largest errors occurred toward the narrower ends.

V. DISCUSSION

The insertion forces measured for the robotically inserted rigid probe (3.22 N) were consistent with values reported for conventional cervical instruments (1.5 to 6.5 N) [15],

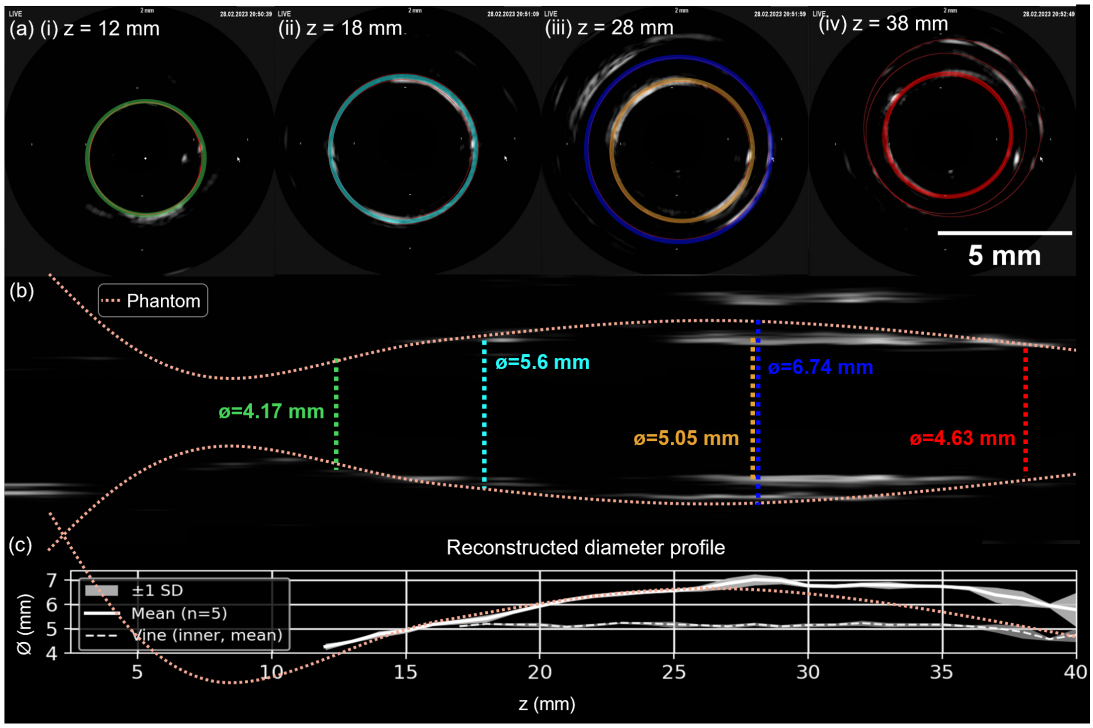


Fig. 5. Ultrasound-based reconstruction of the simulated cervical lumen. (a) Representative radial IVUS slices at insertion depths of (i) 12, (ii) 18, (iii) 28, and (iv) 38 mm. (b) Longitudinal multiplanar reconstruction, correlating the localized luminal diameters with the representative slices. (c) Mean diameter profiles of the vine robot and cervical lumen reconstructed from the IVUS images, with shaded regions denoting ± 1 SD (N=5). The cervical phantom profile is overlaid for reference.

whereas the vine robot eversion required substantially lower forces (0.15 N). These reduced insertion forces measured during phantom navigation, as well as the operating pressure remaining well below the robot's burst limit, support the potential mechanical safety advantages of the proposed soft robotic approach and suggest reduced tissue loading during intervention.

Although the measured data capture the overall geometric trend of the lumen, a slight deviation remains, corresponding to a mean bias of 0.65 ± 0.06 mm with a mean percentage error of $12.8 \pm 1.2\%$ of true diameter. This is likely associated with geometric distortions arising from the use of an IVUS catheter outside its intended vascular imaging application. Moreover, acoustic attenuation and boundary discontinuities caused luminal edges to appear as partial arcs, necessitating manual circular outlining to estimate the complete diameter, which likely exacerbated the error.

Furthermore, despite the use of a robotic system allowing precise motion of the IVUS probe, some error in the effective Δz step size could contribute to an accumulated distance error. This may arise from the vine growth mechanism, in which the tail (in this case the probe) is assumed to advance at twice the speed of the vine body. Although this relationship was considered in the setup, it remains an approximation and may introduce small deviations in the reconstructed spatial position. Consequently, reconstructed cross-sections may correspond to slightly smaller spatial intervals than assumed, resulting in the increased error observed toward the narrower ends of the phantom. Incorporating proprioceptive sensing to monitor growth advancement could reduce uncertainty in the effective probe displacement and improve accuracy of future reconstructions.

Corresponding ultrasound imaging of the ex vivo tissue was limited by high tissue attenuation and reduced contrast, preventing reliable feature identification. These findings suggest that while robotic adaptability for deployment is feasible in biological tissue, real-time ultrasound imaging during navigation requires improvement in acoustic sensitivity and coupling. Nonetheless, these preliminary results demonstrate the feasibility of using a soft vine robot for minimally invasive cervical navigation into the uterus.

VI. CONCLUSION AND FUTURE WORK

In conclusion, this work shows the deployment of a soft vine robot with an integrated IVUS probe for minimally invasive cervical navigation. The ex vivo experiments demonstrate the feasibility of the proposed approach in biological tissue, while the reduced insertion forces measured in the phantom highlight the potential safety advantages of using a soft robot compared with conventional rigid probes. The ultrasound imaging results indicate that the proposed system can acquire spatially meaningful ultrasound data during soft robotic cervical deployment, while highlighting the need for a higher-sensitivity ultrasound probe in future work.

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